WORKED SOLUTION

STEP II 2019, Question 5

Pure • Fixed Points of Iteration

STEP II 2019, Question 5

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

A constant sequence starts at a *fixed point*, a solution of $f(x) = x$. A sequence of period 2 starts at a solution of $f(f(x)) = x$ that is not a fixed point. Every fixed point also solves $f(f(x)) = x$, so $f(x) - x$ must be a factor of the quartic $f(f(x)) - x$. Once we have the other factor, which is a quadratic, everything comes down to discriminants.

Solution

A factorisation for both parts

In both parts $f(x) = x^2 - px + c$ for a constant c ($c = p$ in (i), $c = q$ in (ii)). For any x and y,

$$f(y) - f(x) = (y^2 - x^2) - p(y - x) = (y - x)(y + x - p).$$

Putting $y = f(x)$ and adding $f(x) - x$ to both sides gives

$$\begin{aligned} f(f(x)) - x &= [f(f(x)) - f(x)] + [f(x) - x] \\ &= (f(x) - x)(f(x) + x - p + 1). \end{aligned}$$

Two consequences we shall use:

- The sequence has period 2 exactly when a is a root of $Q(x) = f(x) + x - p + 1$ that is *not* a root of $f(x) - x$. Indeed, if $f(f(a)) = a$ then $u_2 = u_0$, and since each term is determined by the one before, $u_{n+2} = u_n$ for all n by induction. The sequence is then non-constant precisely when $u_1 \neq u_0$, i.e. $f(a) \neq a$.
- If x is a fixed point then $Q(x) = 2x - p + 1$. So a fixed point is also a root of Q only if $x = \frac{1}{2}(p - 1)$.

Part (i)

Here $f(x) = x^2 - px + p$, so

$$f(a) - a = a^2 - (p + 1)a + p = (a - 1)(a - p).$$

The sequence is constant exactly when $u_1 = u_0$ (then every term equals a), so the values are

$$a = 1 \quad \text{and} \quad a = p.$$

For period 2, the other factor is

$$Q(x) = x^2 - px + p + x - p + 1 = x^2 + (1-p)x + 1,$$

with discriminant

$$(1-p)^2 - 4 = (p - 1 - 2)(p - 1 + 2) = (p-3)(p+1).$$

- If $-1 < p < 3$, the discriminant is negative, Q has no real roots, and no real a gives period 2.
- If $p = 3$ or $p = -1$, Q has the single root $x = \frac{1}{2}(p-1)$, which is 1 or -1 respectively. For $p = 3$ the fixed points are 1 and 3; for $p = -1$ they are 1 and -1. In both cases the root of Q is a fixed point, so the sequence starting there is constant, not of period 2.
- If $p > 3$ or $p < -1$, Q has two distinct real roots. Neither is a fixed point: by the second bullet above it would have to be $\frac{1}{2}(p-1)$, but

  $$Q\left(\tfrac{1}{2}(p-1)\right) = \tfrac{1}{4}(p-1)^2 - \tfrac{1}{2}(p-1)^2 + 1 = 1 - \tfrac{1}{4}(p-1)^2 \neq 0$$

  because $(p-1)^2 > 4$. So taking a to be either root gives a sequence of period 2.

Hence a value of a giving period 2 exists if and only if $p > 3$ or $p < -1$.

Part (ii)

Now $\mathrm{f}(x) = x^2 - px + q$, so

$$\mathrm{f}(x) - x = x^2 - (p+1)x + q.$$

A constant sequence needs a with $\mathrm{f}(a) = a$, i.e. a real root of this quadratic.

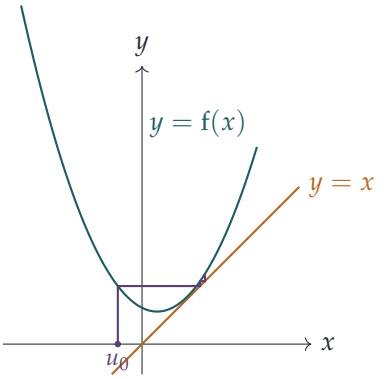
If $\mathrm{f}(x) > x$ *for all* x, then $\mathrm{f}(a) \neq a$ for every a, so no sequence is constant.

Conversely, suppose no a gives a constant sequence, so $g(x) = x^2 - (p+1)x + q$ has no real root. Then g is continuous and never zero, so it never changes sign, and since $g(x) \to +\infty$ as $x \to \infty$ it is positive everywhere. (Equivalently: the discriminant is negative and the leading coefficient positive.) That is, $\mathrm{f}(x) > x$ for all x, which completes the "if and only if".

The deduction. Suppose there is no constant sequence. Then $\mathrm{f}(x) > x$ for all x, so for any a,

$$u_{n+1} = \mathrm{f}(u_n) > u_n \quad \text{for every } n.$$

The sequence is strictly increasing, so $u_2 > u_1 > u_0$. In particular $u_2 \neq u_0$, and the sequence cannot have period 2.



The picture shows the situation: the parabola lies wholly above $y = x$, so the "staircase" of iterates can only climb.

The final part. The converse is *false*. Take $p = q = 0$, so $\mathrm{f}(x) = x^2$. The sequence is constant for $a = 0$ and $a = 1$. For period 2 we need $\mathrm{f}(\mathrm{f}(a)) = a^4 = a$, i.e. $a(a-1)(a^2+a+1) = 0$. The factor $a^2 + a + 1$ has negative discriminant, so the only real solutions are $a = 0$ and $a = 1$, both fixed points. So there is no value of a giving period 2, even though some values give a constant sequence.

> **Ask yourself** Is $\mathrm{f}(x) = x^2$ just a lucky accident?

No. In general the fixed-point quadratic has discriminant $D = (p+1)^2 - 4q$, and $Q(x) = x^2 + (1-p)x + q - p + 1$ has discriminant

$$(1-p)^2 - 4(q-p+1) = (p+1)^2 - 4q - 4 = D - 4.$$

As in (i), a root of Q is a fixed point only when the discriminant of Q is zero. So constant sequences exist when $D \geq 0$, and period-2 sequences exist when $D > 4$. Any p, q with $0 \leq D \leq 4$ gives a counterexample.

Checks

- With $p = 4$ in (i), $Q(x) = x^2 - 3x + 1$ has roots $\frac{1}{2}(3 \pm \sqrt{5}) \approx 2.618,\ 0.382$. Starting at 2.618, $\mathrm{f}(2.618) = 4 + (2.618 - 4)(2.618) \approx 0.382$, and $\mathrm{f}(0.382) \approx 2.618$: period 2, as predicted.
- At $p = 3$ the root of Q is $x = 1$, and $\mathrm{f}(1) = 3 + (1 - 3) = 1$: a fixed point, confirming that the boundary values are excluded.
- Expanding $(x^2 - (p+1)x + p)(x^2 + (1-p)x + 1)$ gives $\mathrm{f}(\mathrm{f}(x)) - x$; a computer check at thousands of random (p, x), together with the condition "period 2 iff $D > 4$" in (ii), found no exceptions.

Common slips

- Including $p = 3$ and $p = -1$: there $\mathrm{f}(\mathrm{f}(a)) = a$ has a repeated root, but it is a fixed point, so the sequence is constant.
- Forgetting that the factor $\mathrm{f}(x) - x$ must be removed from $\mathrm{f}(\mathrm{f}(x)) - x$ before looking for period-2 points.
- In the final part, arguing from the direction of the inequality alone; a single explicit counterexample is what settles a "Is it true that…?".

Beyond STEP

This is the first step into *discrete dynamical systems*. Every quadratic map is equivalent, by a linear change of variable, to the logistic map $x \mapsto rx(1 - x)$. As the parameter grows, the fixed point gives way to a 2-cycle, then a 4-cycle, and so on, in a period-doubling cascade that ends in chaos. The thresholds you found, where the discriminant of Q vanishes, are the first of these bifurcations. Strogatz's *Nonlinear Dynamics and Chaos* is a very readable introduction.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.